

Project Title

OptiCrop – Smart Agricultural Production Optimization Engine

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Introduction

OptiCrop is an intelligent Machine Learning-based web application developed to recommend the most suitable crop for a given plot of land. The system helps farmers, agribusinesses, and agricultural researchers make faster, more accurate, and data-driven planting decisions by analyzing soil and climate readings and predicting the crop most likely to succeed.

The application leverages a curated agricultural dataset and multiple analytical techniques – including K-Means clustering to uncover soil/climate patterns and a Logistic Regression classifier for crop prediction – to identify the most suitable crop for a given input profile. The best-performing model is integrated into a Flask-based web application that enables users to enter soil and climate readings and receive an instant crop recommendation.

The system evaluates various factors such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, soil pH, rainfall, and seasonal patterns before generating a prediction.

To enhance the user experience, the application provides a modern and responsive interface where users can submit their soil and climate readings, view the recommended crop on the same page, and read a short description of why that crop suits their conditions. This feature makes the system more informative and user-friendly by helping farmers understand and act on the recommendation.

Ideation Phase

Brainstorm & Idea Prioritization Template

Field	Value
Date	7/9/2026
Name	K.Hemanth Kumar
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

OptiCrop is a Machine Learning-powered web application designed to recommend the most suitable crop for cultivation based on soil nutrients and climate conditions. The system helps farmers, agribusinesses, and agricultural researchers make faster, more reliable, data-driven planting decisions by analyzing soil and environmental readings and determining which crop is most likely to give the best yield.

The application evaluates soil and climate patterns using clustering and classification algorithms and deploys the best-performing model through a Flask-based web interface for real-time recommendations.

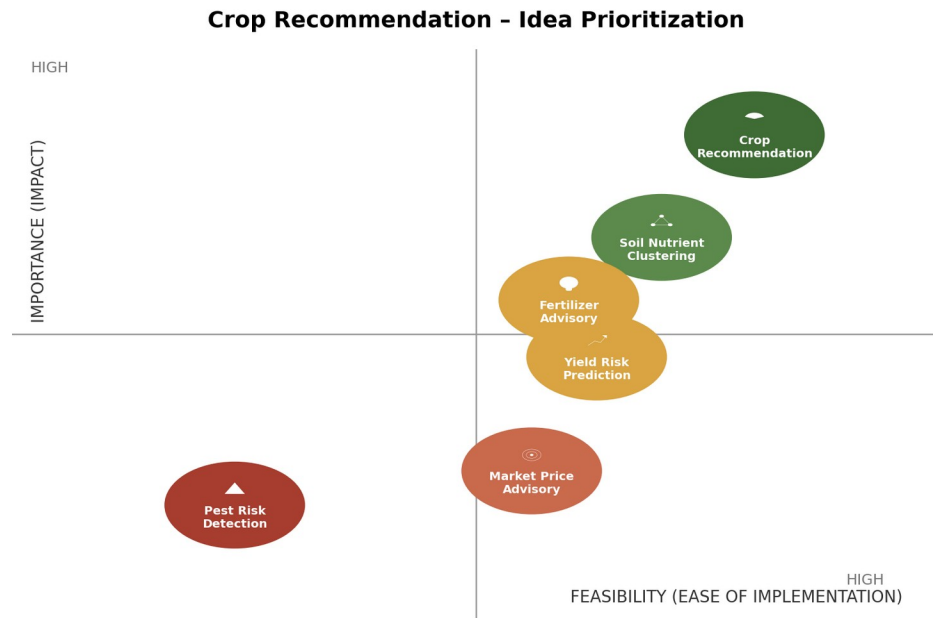
Step-1: Selecting the Problem Statement:

PROBLEM: How might we recommend the most suitable crop for a given soil and climate profile using Machine Learning?

Step-2: Idea Phase

Idea A – Crop Advisory	Idea B – Resource & Yield Optimization
Crop Suitability Prediction	Fertilizer Requirement Estimation
Soil Nutrient Analysis	AI-Based Soil Evaluation
Automated Crop Advisory	Crop Clustering Analysis
Smart Farming Platform	Water Requirement Prediction
Real-Time Crop Recommendation	Seasonal Crop Planning

Step-3: Idea Prioritization



Idea prioritization quadrant for OptiCrop

Ideation Phase

Define the Problem Statements

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Maximum Marks	2 Marks

Customer Problem Statement Template:

 I AM	Who are they?	Farmers, small and marginal landholders, agricultural advisors who need reliable crop guidance.
 I'M TRYING TO	What do they want to achieve?	Choose the crop that will grow best and yield the most on their land, right now.
 BUT	What's getting in their way?	Crop selection is often based on habit or guesswork rather than actual soil and climate conditions.
 BECAUSE	Why is this happening?	Manual soil-report interpretation, lack of real-time data analysis, and limited agronomic expertise.
 WHICH MAKES ME FEEL	How does this make them feel?	Uncertain and financially exposed to crop failure.

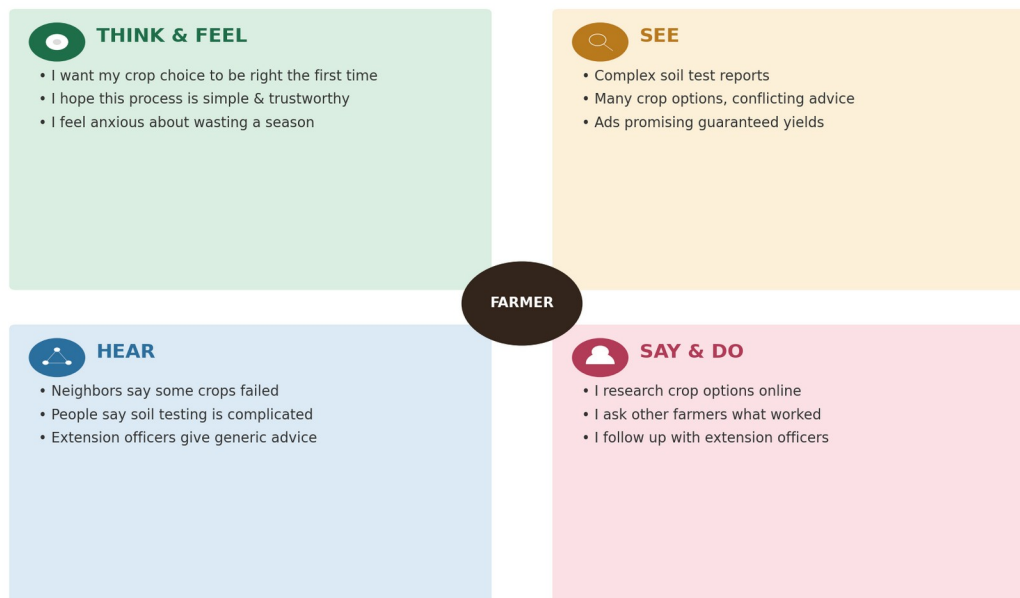
Ideation Phase

Empathize & Discover

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Empathy Map Canvas:

Empathy Map - OptiCrop Crop Recommendation System

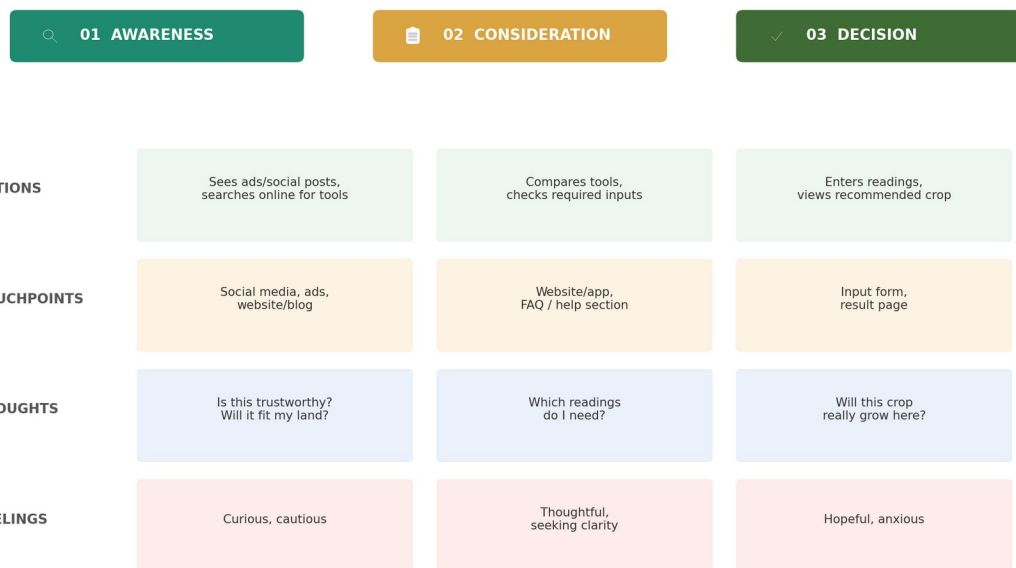




Customer Journey Map

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Customer Journey Map - OptiCrop

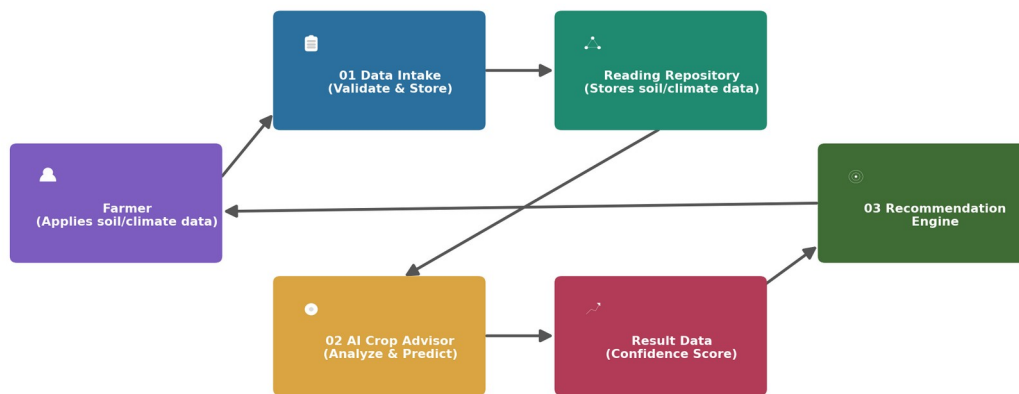


Data Flow Diagram & User Stories

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Maximum Marks	4 Marks

Data Flow Diagrams:

OptiCrop - Smart Agricultural Production Optimization Engine



User stories:

User Type	Epic	US No.	User Story / Task	Priority	Release
Farmer	Soil Data Entry	USN-1	As a user, I can enter my soil and climate readings.	High	Sprint-1
Farmer	Crop Recommendation	USN-2	As a user, I can submit readings and get a crop recommendation.	High	Sprint-1
Farmer	View Crop Info	USN-3	As a user, I can view a short description of the recommended crop.	Medium	Sprint-2
Researcher	Seasonal Insights	USN-5	As a researcher, I can view seasonal crop groupings.	Medium	Sprint-2
Administrator	Model Monitoring	USN-6	As an administrator, I can monitor prediction accuracy.	Medium	Sprint-3



Solution Requirements

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Functional Requirements:

Functional Requirement	Sub Requirement
Soil & Climate Input	Required fields: N, P, K, temperature, humidity, pH, rainfall
Crop Recommendation	View the predicted crop
Crop Information	View a short description of the recommended crop
Navigation	View Home, About, and Find your crop pages

Non- Functional Requirements:

Functional Requirement	Sub Requirement
Usability	Application should provide a simple interface for all users.
Reliability	Application should provide accurate and consistent crop recommendations.
Performance	Crop recommendation results should be generated within a few seconds.
Availability	Application should be available with minimal downtime.
Scalability	Application should support multiple users and increasing requests efficiently.

Technology Stack

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Technical architecture:

The deliverable shall include the architectural diagram as below and the information.

Components	Description	Technology
User interface	Web interface for entering soil/climate data and viewing the recommendation	HTML, CSS
Application logic 1	Page routing and navigation	Python (Flask)
Application logic 2	Soil/climate data validation and preprocessing	Python, Pandas
Application logic 3	AI-based crop recommendation	Scikit-learn, NumPy
Machine learning model	Crop recommendation model	Logistic Regression, K-Means
Infrastructure	Application deployment on server or cloud	Local server, Render

Application characteristics:

Characteristics	Description	Technology
Open source frameworks	List the open-source frameworks used	Flask, Pandas, scikit-learn
Security implementations	Access controls to be used	HTTPS or local host
Availability	Justify the availability of the application	Local / cloud hosting
Performance	Design consideration for performance	Pandas, NumPy, optimized model

Project Design Phase

Solution architecture

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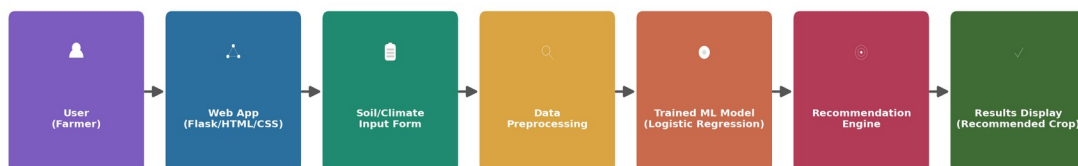
Solution architecture:

Solution architecture is a process of designing a machine learning based web application that connects a farmer's soil and climate inputs with a crop recommendation.

It bridges the gap between crop selection challenges and an intelligent, data-driven software solution.

- It develops a reliable machine learning solution for recommending crops.
- It describes the system architecture, workflow, and functionality used to generate the crop recommendation.
- It provides a structured framework for developing, deploying, managing, and maintaining the crop recommendation system.

Solution Architecture Diagram:



Project design Phase

Proposed solution

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Proposed solution:

Parameter	Description
Problem statement	Farmers often select crops based on habit rather than actual soil and climate suitability, reducing yield.
Idea or solution description	Develop a machine learning based web application that recommends the most suitable crop based on soil nutrients and climate conditions.
Novelty or uniqueness	The system provides instant, data-driven crop recommendations backed by clustering and classification analysis.
Social impact	Helps farmers make faster, more confident planting decisions and supports better use of resources.
Revenue model	The solution can be offered to agri-tech companies, cooperatives, and extension services as a software service.
Scalability of the solution	The application can be scaled to support more crops, larger datasets, and additional regions.

Project Design Phase

Problem - solution fit

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Project solution fit template:

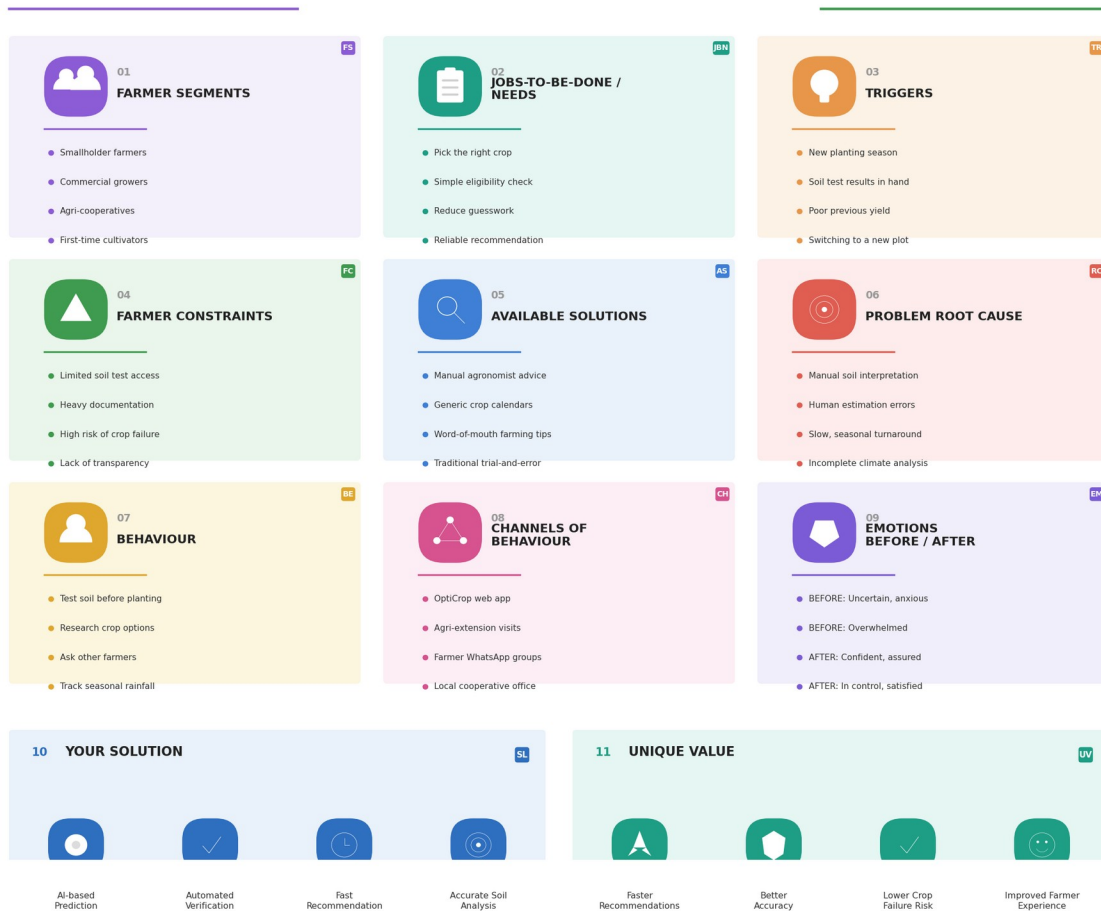
The problem-solution fit means identifying the challenges faced by farmers and agricultural stakeholders in selecting suitable crops, and providing a machine-learning-based solution that accurately recommends the best crop for given soil and climate conditions, reducing guesswork and improving yield outcomes.

Main purpose:

- Reduces reliance on guesswork and improves crop selection accuracy.
- Improves farmer confidence with data-backed recommendations.
- Helps researchers and policymakers analyze crop-environment patterns.
- Enhances resource efficiency for water, fertilizer, and soil nutrients.
- Recommends the crop best suited to a specific plot's nutrient and climate profile.
- Helps farmers plan seasonal crop choices in advance.

CROP RECOMMENDATION PROCESS & OPTIMIZATION FRAMEWORK

A holistic approach to streamline crop selection decisions



Crop Recommendation Process & Optimization Framework



Project Planning Phase

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Maximum Marks	5 Marks

Product Backlog, Sprint Schedule and Estimation:

Sprint	Epic	US No.	Task	Points	Priority
Sprint-1	Data Collection	USN-1	Collect the crop recommendation dataset for model training.	2	High
Sprint-1	Data Preprocessing	USN-2	Clean and preprocess the dataset.	2	High
Sprint-2	Clustering	USN-3	Apply K-Means clustering to identify soil/climate patterns.	3	Medium
Sprint-2	Model Training	USN-4	Train the Logistic Regression model for crop prediction.	3	High
Sprint-3	Flask Integration	USN-6	Integrate the trained model with the Flask application.	3	High
Sprint-4	Crop Recommendation	USN-8	Predict the most suitable crop from user input.	3	High
Sprint-4	Testing & Deployment	USN-9	Test and deploy the application successfully.	3	Medium



Code-Layout, Readability and Reusability

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Maximum Marks	5 Marks

This document evaluates the quality of the code base in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design.

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python, HTML, CSS files follow proper indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into data, model, app, templates, and static folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as FEATURE_COLUMNS, CROP_INFO, and prediction are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly indicate routing, prediction, and preprocessing.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections are documented to improve understanding and maintenance.
6	Modular Design	Code is split into reusable functions/modules	Yes	Dataset generation, training, and the Flask app are separated into different files.
7	No Redundant Code	Duplicate or unused code is removed	Yes	The implementation avoids unnecessary duplicate code and keeps modules concise.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Basic validation and error handling are implemented to manage invalid user input.

Reusable Components:

S.No	Component / Module	Technology Used	Purpose	Reusability
1	Flask Web Application	Flask	Handles user requests and	High



S. No	Component / Module	Technology Used	Purpose	Reusability
	(app.py)		recommendation workflow	
2	Crop Recommendation Model (model.pkl)	Scikit-learn	Used for every crop prediction	High
3	Data Preprocessing Module	Pandas, NumPy	Used before every prediction	High
4	HTML Templates	HTML5	User Interface	Medium
5	Static Resources	CSS3, JavaScript	UI styling and interaction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for data, model, and app.
Readability	5	Code is easy to understand with meaningful names and proper formatting.
Reusability	5	Model, preprocessing logic, and Flask application can be reused or extended easily.
Documentation / Comments	5	Code is documented sufficiently for understanding and future maintenance.
Overall Score	5/5	The project follows a modular, maintainable, and reusable architecture suitable for deployment and future enhancement.

Coding and solution

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Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented code and the overall solution, demonstrating working functionality, clean code practices, and alignment with the defined requirements.

Solution Summary:

Field	Details
Repository Link / URL	https://github.com/hemanthkumar246/Opti_Crop/tree/main
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask, scikit-learn
Key Features Implemented	Crop recommendation using Machine Learning; data preprocessing; Flask-based web application; user-friendly soil/climate input form; real-time crop recommendation; trained model integration using pickle; input validation and result display.
Pending / Incomplete Features	Cloud deployment; database integration; user authentication; recommendation history storage.
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install the required Python libraries using requirements.txt. 3. Open the project folder. 4. Run app.py. 5. Open the Flask application in the browser. 6. Enter soil/climate details and click Predict.

Code Quality Checklist:

S.No	Criteria	Status
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes



No. of Functional Features included in the Solution

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Maximum Marks	5 Marks

Functional Features Overview

Feature Name	Feature Description	Module/ Component	Status	Marks
Soil & Climate Input Form	Allows users to enter N, P, K, temperature, humidity, pH, and rainfall readings.	Frontend (HTML/CSS)	Done	High
Input Validation	Validates user input before processing.	Flask Backend	Done	High
Data Preprocessing	Cleans and transforms input data into the required format.	Preprocessing Module	Done	High
Crop Recommendation	Predicts the most suitable crop using the trained ML model.	Machine Learning Model	Done	High
Result Display	Displays the recommended crop and a short description to the user.	Flask Application	Done	High
Trained Model Loading	Loads the saved machine learning model (model.pkl) for prediction.	Model Loader	Done	High
Responsive User Interface	Provides a user-friendly interface accessible through a web browser.	HTML/CSS/JavaScript	Done	High
Error Handling	Handles invalid or incomplete user input gracefully.	Flask Backend	Done	High

Performance testing

Performance model test

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Maximum Marks	5 Marks

Test scenarios and results:

Test ID	Scenario	Expected Result	Actual Result	Pass/ Fail
FT-01	User Input Validation	Valid inputs accepted; invalid inputs show an error.	Worked as expected.	Pass
FT-02	Crop Recommendation	System predicts the most suitable crop.	Prediction displayed successfully.	Pass
FT-03	Model Integration Check	Application sends input to the ML model and receives a prediction.	Model returned a prediction successfully.	Pass
FT-04	Application Navigation	Pages load correctly without errors.	All pages loaded successfully.	Pass
PT-01	Response Time Test	Prediction displayed within 3 seconds.	Prediction displayed within 3 seconds.	Pass
PT-02	Multiple Prediction Test	Application responds correctly without slowing down.	Multiple predictions completed successfully.	Pass
PT-03	Application Stability Test	Application should work without crashing.	Application remained stable throughout testing.	Pass



Project Executable Files

Field	Value
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Maximum Marks	3 Marks

S.No	Item to Submit	Submitted
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt	Yes
4	Database schema / seed files (if applicable)	N/A
5	Environment configuration (.env.example)	No
6	Deployed application URL	No
7	APK / Executable binary	N/A
8	Dockerfile / Containerization config	No
9	Test files and test results	Yes
10	Demo video / walkthrough	Yes

File / Folder Structure:

OPTICROP/

```

├── data/
│   ├── Crop_recommendation.csv    # Dataset used for training
│   └── generate_dataset.py        # Script that builds the dataset
├── model/
│   ├── train_model.py            # EDA, preprocessing, clustering, training
│   ├── model.pkl                 # Trained Logistic Regression model
│   └── plots/                    # EDA charts + classification report
├── app/
│   ├── app.py                    # Main Flask application
│   ├── static/style.css          # CSS styling
│   └── templates/                # base.html, home.html, about.html, findyourcrop.html

```

```
└─ requirements.txt      # Python dependencies
└─ README.md            # Project documentation
```

Deployment / Access Details:

Field	Value
Hosted / Deployed URL	Not Hosted (Runs Locally)
Login Credentials	Not required – no authentication in this version
Platform / Hosting Provider	Local Flask Server
Repository Link	https://github.com/hemanthkumar246/Opti_Crop/tree/main
Demo Video Link	[project demo video link]

Run Instructions:

- Clone the repository.
- Install dependencies using requirements.txt.
- Run app.py.
- Open <http://127.0.0.1:5000> in your browser.
- Enter soil and climate details and click Predict.

Known Issues / Limitations:

S.No	Known Issue	Workaround / Status
1	Runs locally only	Deploy to cloud
2	No database	Can integrate MySQL/PostgreSQL
3	No user authentication	Can be added later

Communication

Field	Value
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Maximum Marks	1 Marks

Communication Plan:

S. No	Type	Frequency	Channel / Tool	Participants	Purpose
1	Project Planning	Daily	Personal Notes, VS Code	Self	Plan daily development tasks
2	Progress Review	Weekly	GitHub, VS Code	Self	Review completed modules and pending work
3	Issue / Bug Resolution	As Needed	Stack Overflow, Docs, GitHub	Self	Resolve coding and runtime issues
4	Guide / Mentor Discussion	Weekly	Google Meet / WhatsApp	Student & Guide	Discuss progress and receive feedback
5	Documentation Review	Once	Microsoft Word	Self	Verify documentation before submission
6	Final Project Submission	Once	Skill Wallet & GitHub	Self	Submit final project and documentation

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Difficulty integrating the trained crop recommendation model with Flask	Verified file paths, loaded model.pkl correctly, tested predictions locally.
2	Errors due to missing Python libraries	Installed required packages using pip install -r requirements.txt.
3	Preparing documentation according to the SmartBridge template	Updated every section with OptiCrop-specific content, diagrams, and screenshots.



Demonstration of proposed features

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Maximum Marks	1 Marks

Feature Implementation Status:

S.No	Feature Name	Status	Demonstrated	Remarks
1	Crop Recommendation	Implemented	Yes	Prediction generated successfully
2	Soil & Climate Input Form	Implemented	Yes	User-friendly interface
3	Machine Learning Model Integration	Implemented	Yes	Model loaded successfully
4	Responsive Web Interface	Implemented	Yes	Supports desktop and mobile browsers
5	Result Display	Implemented	Yes	Results displayed correctly
6	Project Documentation & GitHub Repository	Implemented	Yes	Repository submitted successfully

Feature Implementation Summary:

Metric	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%

Project Demo Planning

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S. No	Demo Section	Duration	Responsible
1	Introduction – objectives, problem statement, expected outcomes	2 min	Self
2	Solution Overview – architecture, workflow, dataset, ML model	2 min	Self
3	Application Demonstration – live crop recommendation	3 min	Self
4	Feature Explanation – validation, preprocessing, model integration	3 min	Self
5	GitHub Repository & Documentation walkthrough	2 min	Self
6	Conclusion & Future Enhancements	2 min	Self



Scalability & Future Plan

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Maximum Marks	1 Marks

Current System Limitations:

S. No	Limitation	Impact	Priority
1	No user authentication	Unauthorized access may occur	High
2	No database integration	Recommendation history not stored	Medium
3	Limited deployment (local Flask app)	Restricted scalability and accessibility	High

Future Roadmap:

Phase	Planned Feature	Timeline	Expected Impact
Phase 1	Cloud deployment and database integration	1–2 Months	Improved accessibility and data management
Phase 2	User authentication and admin dashboard	2–3 Months	Enhanced security and administration
Phase 3	Recommendation history and analytics dashboard	3–4 Months	Better monitoring and reporting
Phase 4	Advanced ML models and API integration	4–6 Months	Higher accuracy and enterprise scalability

Team Involvement in Demonstration

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Maximum Marks	1 Marks

Team Member	Role	Contribution	Participation
Self (K.Hemanth Kumar)	Presenter & Developer	Developed the project, demonstrated the application, explained the machine learning model, answered questions, and presented documentation.	Active

Aspect	Details
Team Leader / Coordinator	Self (K.Hemanth Kumar)
Overall Team Coordination Rating (1-5)	5/5
Any issues during demo	No major issues encountered.

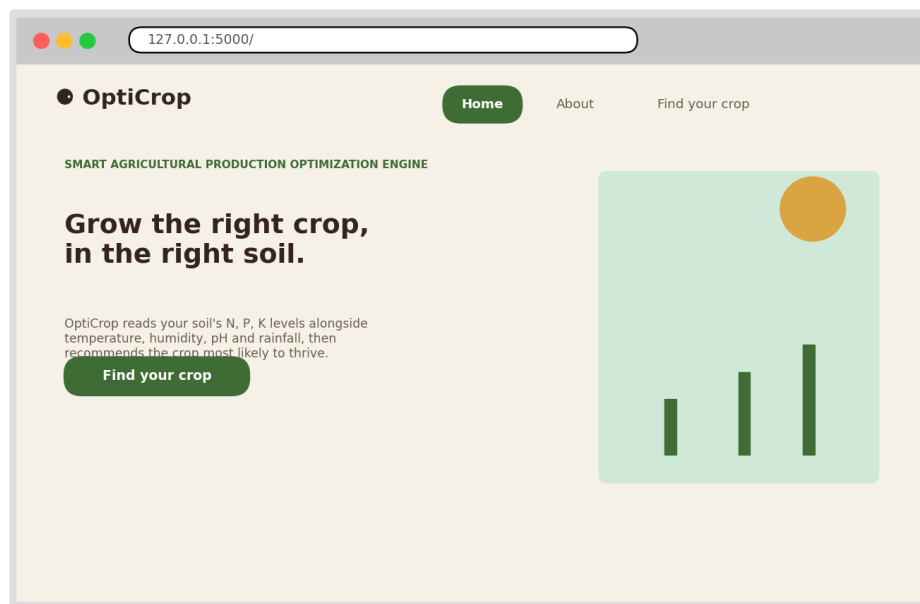
Output Screenshots

OptiCrop's crop recommendation system was successfully developed and tested using a Flask-based web application. The application accepts user input such as Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall. Based on the entered details, the trained machine learning model recommends the crop most likely to succeed.

Note: the previews below are stylized page mockups matching the app's actual design tokens. Swap in real browser screenshots after running the app locally if your submission requires exact captures.

Home page

The home page of the OptiCrop application provides an overview of the crop recommendation system. It introduces the purpose of the application and allows users to navigate to the crop recommendation form by clicking the “Find your crop” button.



Find your crop – input form

The form allows users to enter Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall values. These details are used by the machine learning model to recommend the most suitable crop.



127.0.0.1:5000/findyourcrop

OptiCrop Home About Find your crop

CROP RECOMMENDATION

Find your crop

Nitrogen (N)

Phosphorous (P)

Potassium (K)

Temperature

Humidity

pH

Rainfall

Recommend a crop

Your result will appear here

Crop recommendation result

After submitting valid soil and climate details, the machine learning model analyzes the input and recommends the crop most likely to succeed, along with a short description. This confirms that the application performs real-time crop recommendation successfully.

127.0.0.1:5000/findyourcrop

OptiCrop Home About Find your crop

(soil & climate form filled in)

RECOMMENDED CROP

rice

Thrives in warm, humid conditions with heavy rainfall.

Conclusion

OptiCrop – Smart Agricultural Production Optimization Engine demonstrates how Machine Learning can be effectively applied to improve crop selection decisions in agriculture. By analyzing soil nutrients and climate conditions – Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall – the system accurately recommends the crop best suited to a given plot, enabling faster and more confident planting decisions.

The integration of the trained Logistic Regression model with a Flask-based web application provides a simple, interactive, and user-friendly platform for real-time crop recommendation. The system also groups agricultural conditions using K-Means clustering, offering additional insight into crop-environment relationships.

Overall, the project reduces reliance on habit-based crop selection, minimizes guesswork, improves resource efficiency, and supports farmers and agricultural stakeholders in making data-driven planting decisions. It highlights the practical use of Artificial Intelligence and Machine Learning in solving real-world agricultural problems while enhancing productivity and sustainability. OptiCrop serves as a scalable foundation that can be further enhanced with advanced Machine Learning models, larger and region-specific datasets, cloud deployment, and stronger data management to meet the evolving needs of modern agriculture.